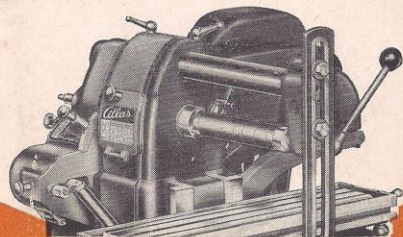
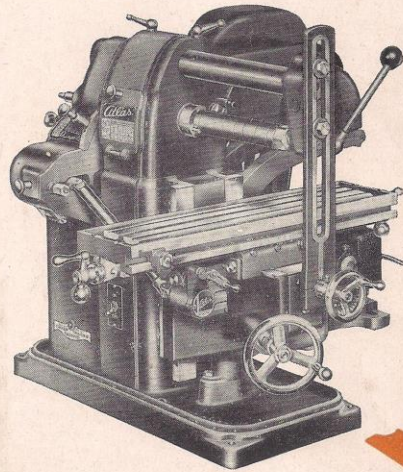
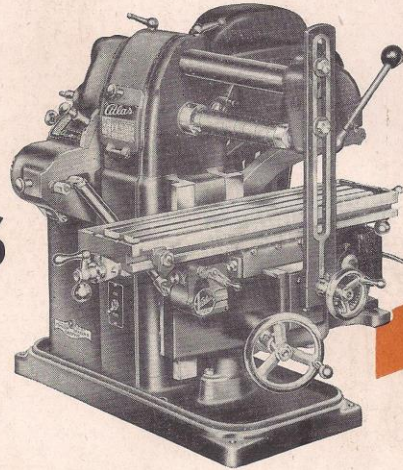
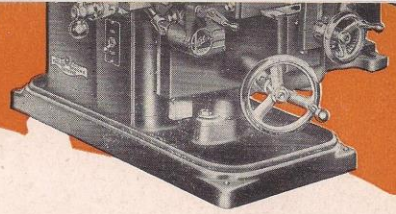


CATALOG NO. M48

Atlas **MILLING MACHINES**



ATLAS PRESS COMPANY

**KALAMAZOO
MICHIGAN
U. S. A.**

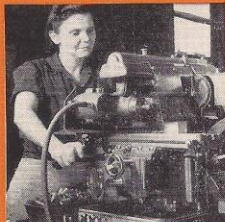
ATLAS IS THE ANSWER TO EFFICIENT MILLING OPERATIONS

RAISED PRODUCTION

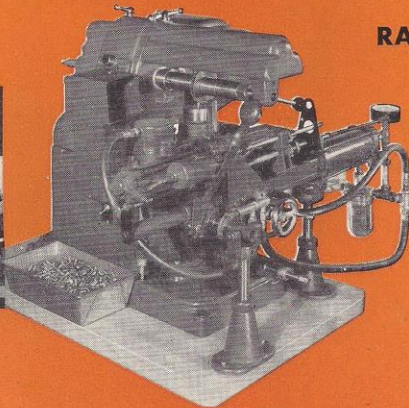
43%

Adding their own air chucks to the standard Atlas miller enabled the Redmer Air Devices Corp., Chicago, to increase production 43%. Their long lived accuracy, rugged strength and adaptability make Atlas millers ideal for the ad-

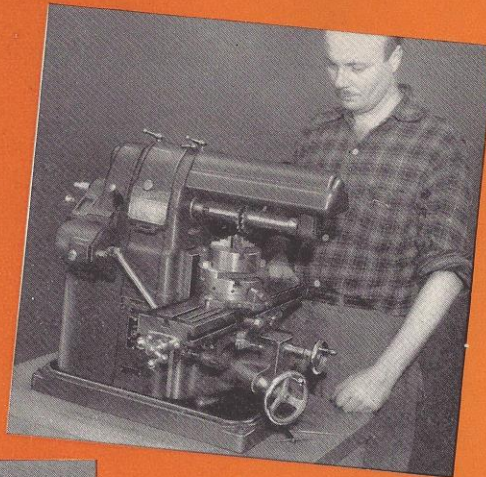
dition of such automatic controls—enabling plant engineers to increase production and cut costs by building semi-automatic machines at only a fraction of the cost of "special" machinery.



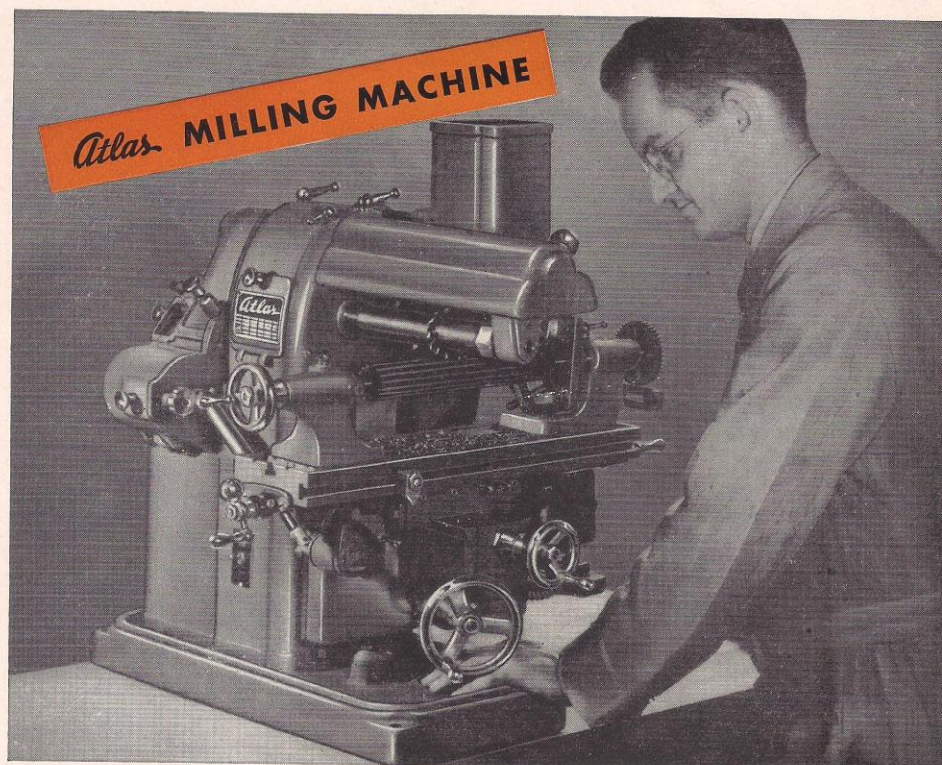
Another view of the specially equipped miller. This installation is typical of the many ways Atlas millers have been adapted to special production problems.



Another important use for Atlas milling machines in many industrial plants is in the research or experimental department, typified by this photograph from the Miller Electro Research Laboratories, Milwaukee. The operator is putting a square end on a round shaft. The part is held by an Atlas universal chuck, which in turn is clamped to an Atlas rotary index table—both standard Atlas accessories.



Atlas milling machines are ideal for school shops. Their compact design gives more pupil stations. Their engineered precision enables students to practice all milling operations to close tolerances, encouraging accurate work habits. They are ruggedly built to stand up under in-expert student use. And advanced Atlas production methods reduce their cost to make limited budgets go farther.



HANDLES EVERY TYPE of MILLING OPERATION

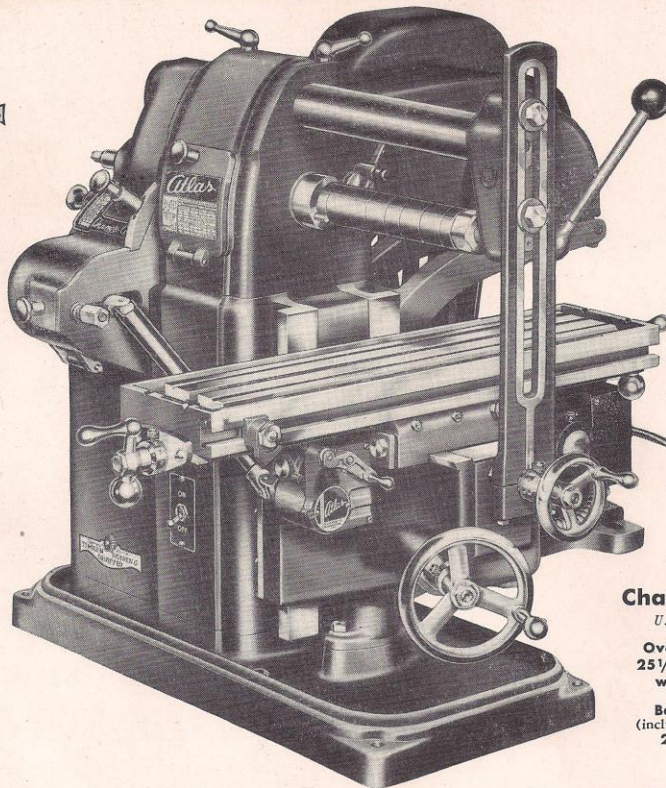
There are spots for this compact milling machine in almost every type of shop — production, tool room, laboratory, maintenance, school and home. It handles the full range of milling, from slabbing and facing cuts to end milling, keyways, finishing, and layout work.

It is ruggedly built for rigid support of cutter and work — precision machined for accuracy in the complete range of milling operations. It has backgeared drive, powered completely by V-belts, plus a Timken bearing equipped spindle, that transmit maximum power to the cutter with a smooth, even action. And it has many easy operating features that reduce set up and operating time.

Three types of Atlas Milling machines are available: the Change-O-Matic, shown above, with reversible automatic power longitudinal feeds; the rapid-production lever control machine; and the manually operated miller. All are fully described on the following pages. Whichever best meets your needs you'll find most economical in original investment — efficient and economical in operation.

- Timken Bearings
- Backgeared Power
- Complete V-Belt Drive
- Eight Spindle Speeds
62 to 2870 RPM
- Precision-Ground
Table 4½" x 18"

Atlas MILLING MACHINE



**No. MFC
Change-O-Matic**
U.S. Patent 2285743

Overall Dimensions
25 1/2" deep x 32 1/2" wide x 22" high

Base Dimensions
(including motor bracket)
20 1/2" x 18 1/2"

COMPLETE SPECIFICATIONS

FEED RANGE	Longitudinal Table Travel.....10" Reversible Automatic Longitudinal Power Feeds: Per Revolution of Spindle.....0.003", .006", .0125", or .025" Range.....0.186" to 11.250" per minute Cross Table Travel, Hand-Operated.....3 3/8" Vertical Table Travel, Hand-Operated.....6" Center of Spindle to Table in Lowest Position.....6"
TABLE	Working Surface, Precision Ground.....4 1/2" x 18" Overall Dimensions.....4 1/2" x 18" x 1 3/8" thick T-Slots.....Top, 3/8" x 5/8" x 16"; Side, 3/8" x 5/8" x 18"
SPINDLE & ARBOR	Eight Spindle Speeds Between.....62 and 2870 RPM Timken Tapered Roller Bearings. Spindle.....1" diameter, 10-pitch N.S. threads, 17/32" hole through entire length, nose bored for No. 2 Morse Taper Arbor Diameter.....7/8" or 1" (see page 10) Center of Arbor to Underside of Overarm.....2 1/8"
V-BELT DRIVE UNIT	Backgear Ratio (approximate).....6 1/2 to 1 Backgear shaft and countershaft spindle bearings.....Oilite Bronze Motor recommended.....1/3 HP 1725 RPM (page 11)

Change-O-Matic TABLE CONTROL

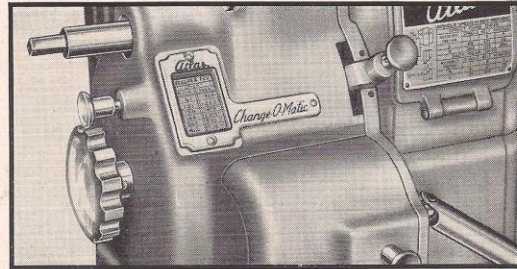
Speeding up all operations requiring longitudinal table travel the exclusive Atlas "Change-O-Matic" mechanism, on MFC machines, gives you instant selection of reversible automatic table feeds between 0.186" and 11.250" per minute. The controls are pictured at the right.

Four automatic feeds are available for each of five spindle speeds—twenty table feeds in all—an ample range for every type of milling work and cutter.

Feeds are selected by turning the large notched knob until desired feed appears through window on "Change-O-Matic" dial chart. The four feed changes (.003", .006", .0125" or .025" per revolution of spindle) list the equivalent in inches per minute for each of five spindle speeds. Table at right summarizes data appearing on four sections of Change-O-Matic chart.

In operating the Atlas, feed direction is reversed instantly by shifting a tumbler lever (see photo) which also has a "neutral" position for disengaging the power feed for hand feeding. Dog in side T-slot of table can be set to stop power feed automatically at any point by tripping a lever on the table feed gear case.

In every respect you will find the Atlas "Change-O-Matic" engineered for efficiency in every milling operation.



The "CHANGE-O-MATIC"

Close-up view of MFC Atlas milling machine showing "Change-O-Matic" controls. Notice (1) large notched knob and index plunger for selecting feed; (2) chart window showing feed selected; (3) tumbler lever controlling direction of table travel; (4) feed gear case; and (5) portion of telescoping drive bar to the table feed screw.

Change-O-Matic TABLE FEEDS (Inches per Minute)

SPINDLE SPEED R. P. M.	FEED PER REVOLUTION OF SPINDLE			
	.003"	.006"	.0125"	.025"
62	.186"	.372"	.775"	1.550"
130	.390"	.780"	1.625"	3.250"
212	.636"	1.272"	2.650"	5.300"
440	1.320"	2.640"	5.500"	11.000"
450	1.350"	2.700"	5.625"	11.250"

This table summarizes data appearing on the Change-O-Matic chart, listing 20 different table feeds in inches per minute.

No. MFC ATLAS MILLING MACHINE complete with safety belt guards and "Change-O-Matic" power feeds, less arbor and motor. Code ZEVSY, net weight 205 pounds, shipping weight 278 pounds.

EQUIPMENT FURNISHED

Atlas "Change-O-Matic" Controls for Instant Selection of Reversible Automatic Table Feeds; Integral V-belt Countershaft; Belts and Pulleys for Complete V-belt Drive; Motor pulley furnished is for 3/8" diameter motor shaft; Motor Mounting Bracket; 12-Ampere Motor Control Switch and Cord—switch is for single phase current only, 3-phase switch is No. S7-300 (page 11); Operating Instructions; less arbor and motor.

CUTTER ARBOR (see page 10)

No. M1-560 CUTTER ARBOR for Atlas milling machine, complete with nut, 6 collars and driver. Code ZETYR. Wt. 5 lb.

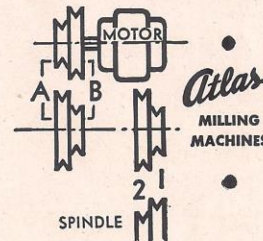
No. M1-576 DRAW-IN BAR. Code word ZEVAM. Weight 1 1/2 lbs.

No. S7-100 CRANK HANDLE for No. M1-576. Code ZEVSYS, Wt. 1 1/2 lbs.

MOTOR RECOMMENDED (see page 11)

The modern efficient Atlas drive permits the economy of a 1/3 HP, 1725 RPM motor. See page 11.

SPINDLE SPEEDS



SPINDLE SPEED IN REV. PER MIN.		
DIRECT CONE DRIVE		
MOTOR BELT POSITION	SPINDLE BELT POSITION	
	1	2
A	450	870
B	1400	2870
BACK GEAR DRIVE		
MOTOR BELT POSITION	SPINDLE BELT POSITION	
	1	2
A	62	130
B	212	440

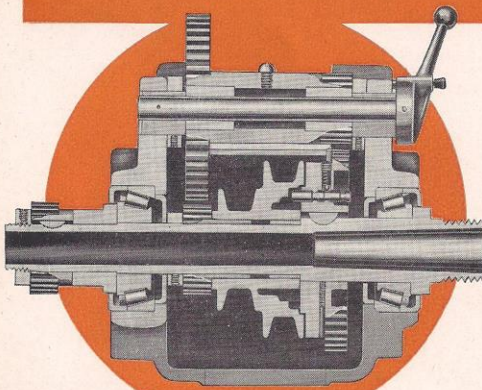
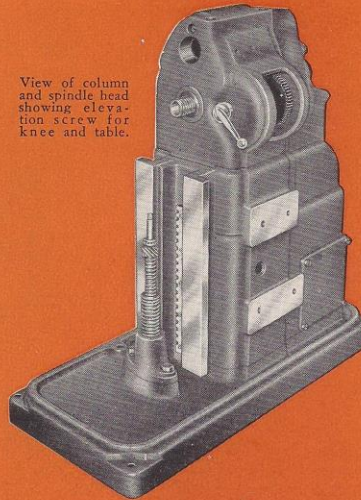
This chart, located on the left side of the Atlas milling machine, shows belt set-ups for eight spindle speeds.

SOUNDLY ENGINEERED FOR PRECISION

MASSIVE COLUMN, HEAD and BASE

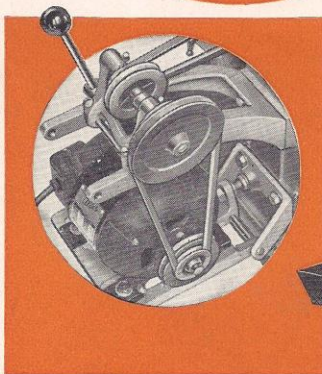
This thick-walled grey iron column (right) firmly supports the spindle head, knee, and countershaft. The ways for the knee bearings, each $1\frac{1}{2}$ " wide and $\frac{1}{2}$ " thick, are cast integrally with the column and precision ground for accurate knee and table alignment and smooth action. Base is a broad and heavy foundation for the entire machine — portion below column forms a reservoir for automatic coolant system (page 11). Spindle head is a heavy well-braced iron casting anchored rigidly to column — a rugged support for spindle and overarm. Accurate line boring of bearing bosses for spindle bearings, backgear shaft, and overarm insures positive alignment. A single phase on-off switch (12 amperes at 125 volts) is built into the column casting.

View of column and spindle head showing elevation screw for knee and table.



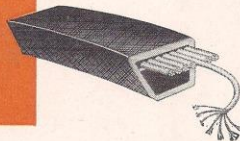
CAREFULLY ENGINEERED SPINDLE AND BEARINGS

Some important reasons for the Atlas miller's power and precision are shown at the left. The spindle is turned from a solid bar of special fine-grained steel and accurately ground. Timken tapered roller bearings maintain accurate spindle alignment, carry all radial and thrust loads. Simple take-up adjustment is provided. These fine bearings permit the wide range of spindle and surface speeds essential for all types of work and cutters. Backgeared drive, powered completely by V-belts, reduces the spindle speed and provides the power for heavy work.



INTEGRAL V-BELT COUNTERSHAFT

The modern V-belt drive transmits maximum power to the spindle with a smooth even action at all eight speeds — 62, 130, 212, 440, 450, 870, 1400 and 2870 RPM. Countershaft is quick change type with belt tension lever in easy reach for speed changes. Lever has two positions — forward to release belts, and back to engage them. Countershaft spindle turns on Oilite bearings. Easy-to-read chart lists spindle speeds for the various belt positions (see page 5). V-belts are highest quality rubber reinforced with endless steel cables. These fine belts assure full transmission of power, resist stretch, last longer. Countershaft and motor base brackets are attached directly to column, making the Atlas milling machine a compact, self-contained unit.



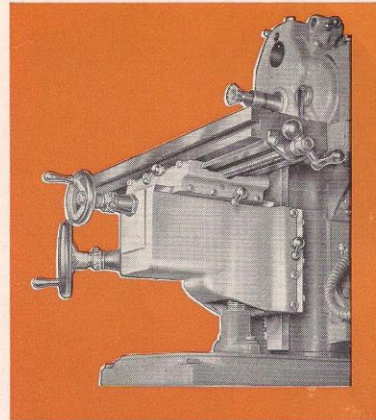
Cross section of Atlas V-belt. Note load-carrying endless steel cable construction.

and POWER on ALL MILLING WORK

RUGGED TABLE SUPPORT

The heavy knee casting is supported by the two $5\frac{1}{2}$ "-long bearings on the column ways and by the telescoping elevation screw. Top of knee has an accurately machined dovetail for the saddle. Column bearing plate with laminated shims and an adjustable gib maintains a firm fit between knee and column ways. Gib lock permits positioning knee rigidly at any height.

The heavy saddle casting has dovetail bearings 5" long on the knee and 6" long on the table. Dovetail ways are accurately machined and hand fitted. Bearings have adjustable gibs. Gib lock permits positioning saddle securely at any point on knee.

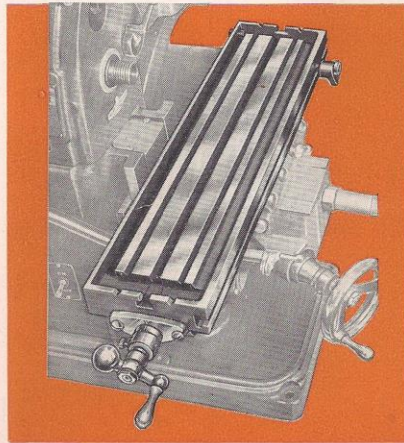


PRECISION GROUND TABLE

The table is a solid grey iron casting 18" long, $4\frac{1}{2}$ " wide and $1\frac{3}{8}$ " thick — a massive work support.

All table surfaces are accurately machined, and top and sides are precision ground. The table top has a T-slot in the center for bolting work and attachments — a V-groove on each side of T-slot simplifies the firm gripping of bar stock. An adjustable dog-type stop in the front table T-slot can be set to stop longitudinal table travel at any point.

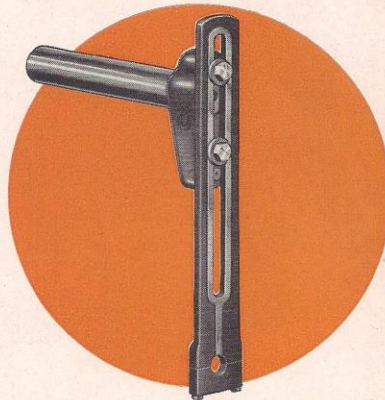
Table dovetail bearing on the saddle is accurately machined and carefully fitted. This bearing is equipped with an adjustable gib — gib screws have lock nuts. A gib lock sets table rigidly at any point on saddle.



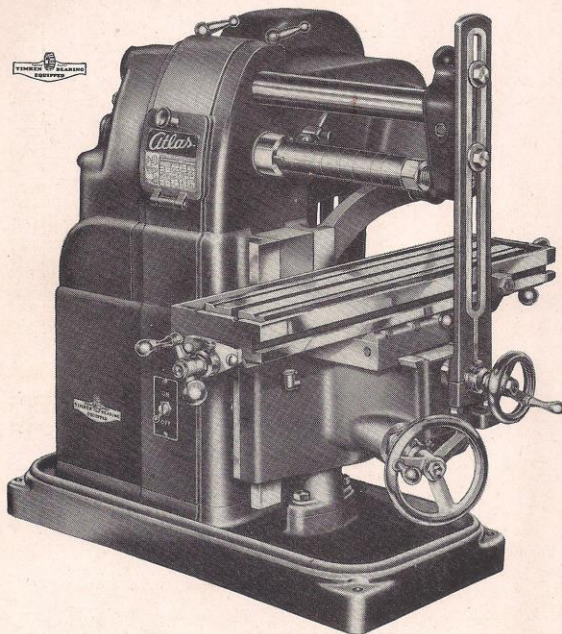
RUGGED OVERARM and ARBOR SUPPORT

Overarm, support bracket and outboard arbor support insure accurate work by maintaining the rigid horizontal position of arbor and cutters. Overarm is a solid bar of stressproof steel, $1\frac{1}{2}$ " in diameter. Bearing bosses for overarm, on each end of spindle head casting, are accurately line bored for positive alignment with spindle. Two coordinate-type clamps lock the overarm rigidly in spindle head. Support bracket is grey iron — has Oilite bronze bearing for arbor. Overarm and support bracket can be removed easily.

Outboard arbor support locks arbor, overarm and knee together rigidly — a heavy malleable iron casting, slotted and machined for quick table height adjustment. Easy to remove and mount.



Atlas MILLING MACHINES



Atlas MILLING MACHINE with MANUAL CONTROLS

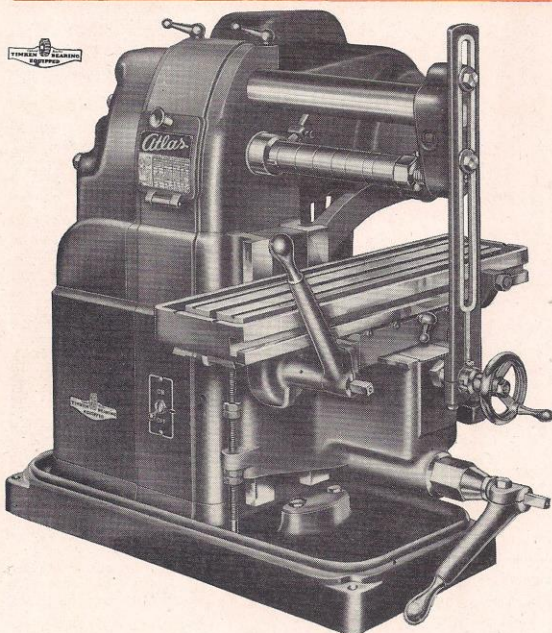
M1C Atlas Milling Machine is equipped with hand-operated screw feeds for table movement. Large front handwheel operates knee and table vertically. Small front handwheel controls cross feed. Ball crank at each end of table permits maximum longitudinal travel of 12". Accurate feed gauging is assured by the feed screw collars, micrometer-graduated in .001ths, with a knurled screw for setting O position at the witness mark. Dog-type stop in front table T-slot can be set to stop longitudinal table travel at any point.

No. M1C ATLAS MILLING MACHINE, complete with safety belt guards and hand-operated table controls, less arbor and motor. Code word ZEVNE, net weight 194 pounds, shipping weight 267 pounds.

FEED RANGE

Longitudinal table travel, hand-operated.....	12"
Cross table travel, hand-operated.....	3 1/2"
Vertical table travel, hand-operated.....	8"
Center of spindle to table in lowest position	6"

NOTE: Specifications, construction and equipment furnished on these models are the same as for the No. M1C described on pages 4, 6 and 7 except for feed range and features described above.



Atlas MILLING MACHINE with PRODUCTION LEVER CONTROLS

The quick-acting lever controls on the MHC Atlas milling machine increase production by speeding up both longitudinal and vertical table movements.

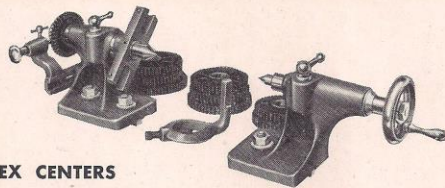
One setting of the left lever permits 5 1/2" longitudinal travel, while second lever moves table 2" vertically at one setting. Each lever operates a rack-and-pinion feed. Lever shafts have squared ends for quick table positioning. Longitudinal travel is set with a dog-type stop in front table T-slot. Vertical travel is fixed by two sets of lock nuts on a screw anchored to base casting.

No. MHC ATLAS MILLING MACHINE, complete with safety belt guards and rapid-production lever controls, less arbor and motor. Code word ZEVUR, net weight 202 pounds, shipping weight 275 pounds.

FEED RANGE

Max. long. travel, 1 setting, lever-operated	4 1/2"
Cross table travel, hand-operated.....	3 3/8"
Max. vertical travel, 1 setting, lever-operated	1 1/2"
Center of spindle to table in lowest position	6"

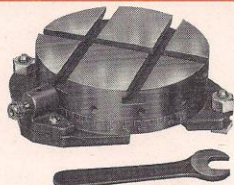
Atlas MILLING MACHINES



INDEX CENTERS

For dividing operations. Headstock and tailstock lock to T-slot in table—keys maintain alignment with T-slot. Centers position for work up to 9" long, lock with coordinate clamps. Maximum swing, 6". Headstock spindle nose has 1" dia., 10-pitch N.S. threads. 17/32" hole thru spindle. (See page 11 for chucks threaded to fit the headstock spindle nose.)

No. M1-200 INDEX CENTERS for Atlas milling machine. Code ZESLE, wt. 15 lb. Furnished: 9 indexing gears (36, 40, 44, 46, 48, 52, 54, 56 and 64 teeth); 1 1/2" dog, driver, bolts; two 60° centers—No. 2 Morse taper for headstock, No. 1 MT for tailstock.

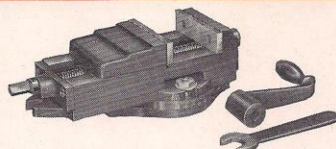
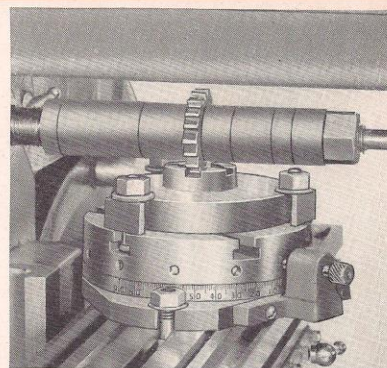


ROTARY INDEX TABLE

Holds work for accurate angular indexing and spacing operations. Table is precision ground on all surfaces, can be swiveled to any angle and locked to base. 3 T-slots in top of table for positioning and locking the work. Swivel base is graduated from 0 to 90° left and right. Indexing holes every 30°.

No. M1-350 ROTARY INDEX TABLE for Atlas milling machines, complete with two base-clamping bolts and wrench. Code ZESNO, wt. 15 lbs.

Table Diameter.....5 1/2" Height to Top of Table.....17 1/8"
Three T-Slots, Top of Table.....3/8" x 5/8"



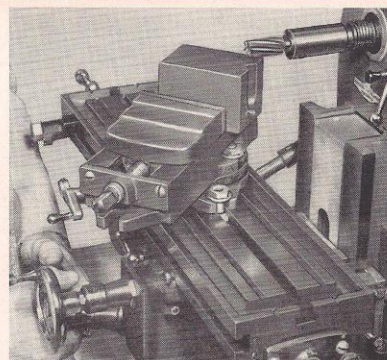
SWIVEL VISE

Grips work rigidly. Vise and base are heavy, machined castings—overall height 3 1/8".

Machined bolt slot at each end and long key in bottom maintain alignment with table T-slot.

Vise turns complete circle—locks in any position—base graduated 0-90° left and right. Jaws are 3" wide, 7/8" high, open 3 1/8"—operated by crank handle furnished.

No. M1-300 SWIVEL VISE with bolts. Code ZESON, weight 15 lbs.

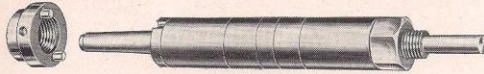


ANGLE PLATE

The angle plate holds irregular shapes that cannot be gripped in vise or bolted to table. A solid well-braced iron casting, accurately machined to a right angle and precision ground on the outer faces. Base and face each measure 2 1/2" x 4" x 7 1/16".

No. S7-430 ANGLE PLATE with bolts. Code ZEHUD, wt. 3 lbs.

ATTACHMENTS and ACCESSORIES FOR

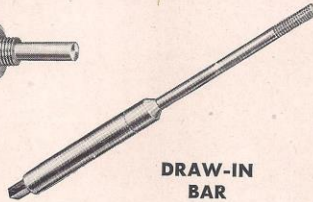


CUTTER ARBORS

Ground stressproof steel with full length keyway and No. 2 Morse taper shank. Driver threaded for spindle nose. Collars $1\frac{3}{8}$ " diameter, precision-ground—two 2", one 1", one $\frac{3}{4}$ ", one $\frac{1}{2}$ ", one $\frac{1}{4}$ " furnished. Length, shoulder to nut, $6\frac{1}{4}$ ". No. M1-576 draw-in bar (right) is required.

No. M1-560 $\frac{7}{8}$ " diam. CUTTER ARBOR for Atlas milling machine, complete with nut, six collars and driver. Code ZETYR, wt. 5 lbs.

No. M1-560L 1" diam. CUTTER ARBOR for Atlas milling machine, complete with nut, six collars and driver. Code ZEWAH, wt. 6 lbs.



DRAW-IN BAR

Required to hold M1-560 or M1-560L cutter arbors, and M1-577 shank cutter adapter, in spindle taper.

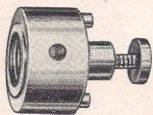
No. M1-576 DRAW-IN BAR, required for cutter arbor and shank cutter adapter. Code ZEVAM, wt. $1\frac{1}{2}$ lbs.



CRANK HANDLE

Required for No. M1-576 draw-in bar and No. M1-300 vise. Drop forged—has accurately broached hole.

No. 57-100 CRANK HANDLE. Code ZEVYS, wt. $1\frac{1}{2}$ lbs.



SHELL END MILL DRIVER

Required for No. M1-585 shell end mill. Is threaded for spindle nose and has two driving pins for slot in back of cutter, ground pilot for cutter hole, large retainer screw.

No. M1-570 SHELL END MILL DRIVER. Code ZEVMA, wt. 2 lbs.



ARBORS FOR ANGULAR CUTTERS

Required for holding No. 574 angular cutters in No. M1-577 shank-cutter adapter. Wt. 8 oz. each.

No.	For No.	Thread	Code
572	574A	24	YEWTE
567	574B	20	YEWUX



SHANK-CUTTER ADAPTER

Holds $\frac{1}{2}$ " straight shank cutters, (angular cutters, end mills, Woodruff keyway cutters). Held in milling machine spindle with M1-576 draw-in bar (required).

No. M1-577 SHANK-CUTTER ADAPTER. Code word ZEVEN, wt. 1 lb.



ANGULAR CUTTERS

For face-milling and dovetailing. Threaded hole—adapted to No. M1-577 shank-cutter adapter with arbors listed above. Wt. 6 oz. each.

No.	Thick	Dia.	Hole	Code
574A	$7/16$ "	$1\frac{1}{4}$ "	$\frac{3}{8}$ "	YALIF
574B	$9/16$ "	$1\frac{1}{2}$ "	$\frac{1}{2}$ "	YALJE



R.H. SPIRAL SHELL END MILL

For wide facing cuts, surfacing, and end milling. Has R.H. spiral teeth. Back is slotted to fit driving pins of M1-570 driver (required). Wt. 2 lbs.

No.	Dia.	Thick	Hole	Code
M1-585	2"	$1\frac{3}{8}$ "	$\frac{3}{4}$ "	ZEVIP



END MILL BUSHINGS

Required to adapt all 576 end mills (except No. 576E) to No. M1-577 shank-cutter adapter. Wt. 4 oz. each.

No. 563E Set of 4 End Mill Bushings. Code YALEB.



WOODRUFF KEYWAY CUTTERS

For cutting Woodruff keyways, slots, grooves, T-slots, etc. Straight shank—held directly in No. M1-577 shank-cutter adapter, 6 oz. each.

No.	Diam.	Thick	Code
575A	$\frac{1}{2}$ "	$\frac{1}{8}$ "	YALUH
575B	$\frac{3}{4}$ "	$3/16$ "	YALYJ
575C	1"	$\frac{1}{4}$ "	YAMAD
575D	$1\frac{1}{8}$ "	$5/16$ "	YAMDA
575E	$1\frac{1}{4}$ "	$\frac{3}{8}$ "	YAMEF



R.H. SPIRAL END MILLS

For slots, straight keyways, facing, routing, squaring and splining shafts. Straight shank—held in No. M1-577 shank-cutter adapter with bushings above. Wt. 4 oz. each.

No.	Length Flute	Diam.	Code
576A	$\frac{5}{8}$ "	$\frac{1}{4}$ "	YAKCE
576B	$11/16$ "	$5/16$ "	YAKEC
576C	$\frac{3}{4}$ "	$\frac{3}{8}$ "	YAKFO
576D	$\frac{7}{8}$ "	$7/16$ "	YAKID
576E	$15/16$ "	$\frac{1}{2}$ "	YAKOF



METAL SLITTING SAWS

Sides taper-ground for proper clearance. Held on M1-560 arbor. $\frac{7}{8}$ " hole. Wt. 8 oz. each.

No.	Diam. ness	Thick	Code
M1-582	$2\frac{1}{2}$ "	$1/32$ "	ZETOP
M1-583	$2\frac{1}{2}$ "	$1/16$ "	ZETPO
M1-584	$2\frac{1}{2}$ "	$1/8$ "	ZETRY



SIDE MILLING CUTTER

Teeth have cutting edges on three sides. Held on M1-560 arbor.

No. M1-581 $2\frac{1}{2}$ " diam., $\frac{1}{4}$ " face, $\frac{7}{8}$ " hole. Code ZETME, wt. 1 lb.

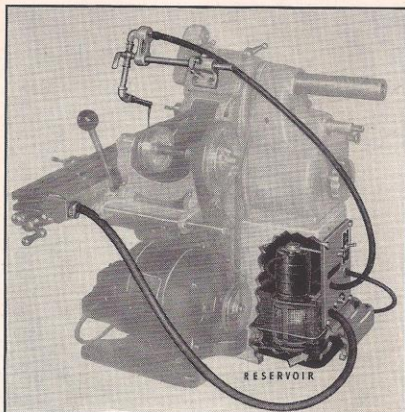


SLAB MILLING CUTTER

For plain surfacing—has R.H. spiral teeth. Held on M1-560 arbor. Hole $\frac{7}{8}$ ". Wt. 2 lbs.

No. M1-580 $2\frac{1}{4}$ " diam., 1" face. Code ZETLA.

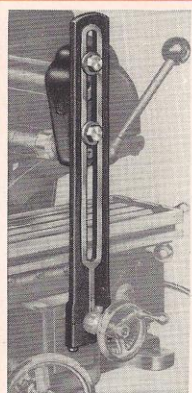
Atlas MILLING MACHINES



PUMPING CAPACITY

(Gallons per Hour)

	Soluble Oil (50% Solution)	Lard Oil (70° F.)
No. M1-600A	198	78
No. M1-600B	360	108



OUTBOARD ARBOR SUPPORT

Locks arbor, overarm and knee together rigidly—improves work finish on heavy cuts by reducing chatter and vibration to a minimum. Fits any Atlas milling machine. Quickly and easily installed or removed—bolts to overarm support bracket and cross feed screw bearing (furnished). Support bar is heavy iron casting, slotted and machined for quick height adjustment.

No. M1-650 OUTBOARD ARBOR SUPPORT for Atlas MF series milling machines. 7 lbs. ZIACY.

No. M1-660 OUTBOARD ARBOR SUPPORT for Atlas M1 and MH series milling machines. 7 lbs. ZIADZ.

Series numbers are shown on nomenclature plate.

MOTORS

No.	Phase	Volt	Weight	Code	Shaft
2730	One	110/220	32	YIDFA	3/8", Single End
2830	Three	220	26	YIDUK	3/8", Single End
2834	Three	440	26	YIEBD	3/8", Single End

All motors listed above are 1/3 HP, 1725 RPM, 60 cycle.

No. 57-300 Three Phase Switch. 6 lbs. ZEBAR.

No. 2710 Cord and Plug for single-phase motor. 1 lb. YIDEG.

AUTOMATIC COOLANT SYSTEMS

These coolant systems, designed especially for use with Atlas milling machines, reduce tool wear, improve work finish, and meet peak production demands. Their simple construction assures smooth, trouble-free performance and long service life.

The modern centrifugal-type pump—engineered for continuous service with any cutting fluid—is installed in 3-pint reservoir compartment in base of the Atlas miller. The coolant stream can be easily directed to any point on cutter with the adjustable outlet nozzle and a sliding arm that supports the rubber feed line. Control valve regulates flow—from slow drip to fast washing action. No priming necessary. Built-in switch.

No. M1-600A COOLANT SYSTEM complete as shown. Motor: induction, 3450 RPM for 110/120 volt, 50-60 cycle AC. Code word ZEJBE, wt. 12 lbs.

No. M1-600B COOLANT SYSTEM for Atlas Milling machines complete as shown. Motor: universal, 6000 RPM for 110/120 volt AC/DC. Code word ZEKEC, wt. 19 1/2 lbs.

COOLANT TANK

This gravity-flow tank is a simple, inexpensive coolant device. Capacity approx. 1 qt. Sliding support arm and adjustable nozzle to direct coolant stream. Valve regulates flow. Nozzle has oil-proof extruded plastic tip.

No. M1-550 COOLANT TANK. Code ZETAL, wt. 8 lbs.



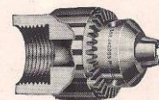
CHUCKS

These chucks are threaded to fit the spindle of Atlas milling machines, the headstock spindle nose of the No. M1-200 index centers (page 9), and the spindle nose of Atlas 6" lathes.

JACOBS CHUCK

For convenient, accurate holding of small work. Handles 1/2" shafts through spindle. Heat-treated steel jaws and body.

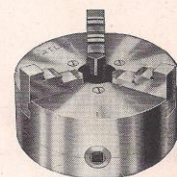
No. M6-375 JACOBS CHUCK capacity No. 70 to 17/32", complete with key-type wrench. YEEBY, 3 lbs.



UNIVERSAL CHUCK

Self-centering jaws controlled by turning one screw. Handles 1/2" rods thru spindle or M1-200 index center. Heat-treated alloy steel jaws, raised and ground steps. High strength semi-steel body.

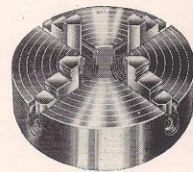
No. M6-845B 4" 3-JAW UNIVERSAL CHUCK complete with 2 sets of jaws (inside and outside) and wrench. Code YIAJK, wt. 5 lbs.



INDEPENDENT CHUCK

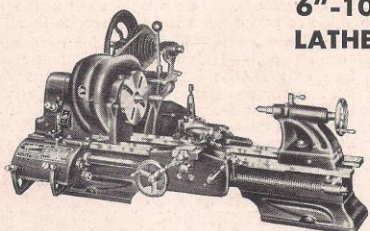
One-piece body of high-strength semi-steel, with face and outer edge ground. Independent, reversible jaws have raised and ground steps. Screws have socket-head for wrench.

No. M6-844B 4" INDEPENDENT CHUCK. Wrench furnished. Code YIALM, weight 6 lbs.

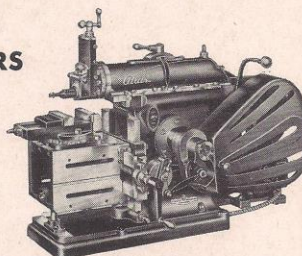


THE *Atlas* 4 TOOL TEAM

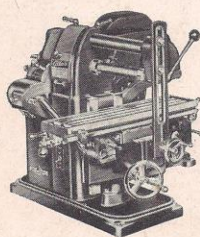
FOR SMALL PARTS MACHINING



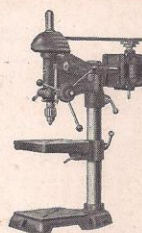
**6"-10"
LATHES**



**7"
SHAPERS**



**MILLING
MACHINES**



DRILL PRESSES

Whatever your use and need for machine tools, there are outstanding values for you in the complete Atlas line — backed by 36 years of precision production experience.

Easy to set up and operate . . . rugged, accurate and efficient . . . Atlas lathes, drill presses, milling machines and shapers have long been outstanding favorites

throughout the world in almost every type of shop in every type of enterprise. Full use of all the latest technological advances in materials, plus unique Atlas-developed production methods, combine to bring these precision tools to you at low cost. Ask your Atlas distributor for complete specifications and prices on the tools that interest you.



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